

Analysis Report

Hospital Patient & Billing Analytics Dashboard

1. PROJECT OVERVIEW

Project: Hospital Patient & Billing Analytics Dashboard

Objective: Analyze patient admission, treatment, and billing data to help hospital management monitor patient trends, admission patterns, and revenue performance.

Dataset: Kaggle “Healthcare Dataset” (synthetic), 55,500 raw patient-level admission records, cleaned to 54,966 rows across 16 columns.

2. DATASET SUMMARY

- Original dataset: 55,500 rows × 15 columns, no missing values in any column
- 534 exact duplicate rows identified and removed
- 108 rows contained negative Billing Amount values (invalid for a billing field) corrected to absolute value
- Date of Admission and Discharge Date were stored as text and converted to proper date type
- Name field had inconsistent capitalization, standardized to proper case
- A derived Length of Stay (days) column was added to support KPI calculations
- Final cleaned dataset: 54,966 rows × 16 columns

3. COLUMN-WISE ASSESSMENT SUMMARY

Full column-by-column detail is available in [Data/assessment/Column_Assessment.xlsx](#).

Column	Type	Issue Found	Action Taken
Name	Text	Inconsistent casing	Standardized to proper case
Date of Admission	Date	Stored as text	Converted to date type
Discharge Date	Date	Stored as text	Converted to date type
Billing Amount	Numeric	108 negative values	Converted to absolute value
(all columns)	—	534 duplicate rows	Removed
Hospital / Doctor	Text	Near-unique per row (~40k distinct)	Kept as descriptive attributes, not modeled as dimensions

4. DATA MODEL OVERVIEW

The model uses a single fact table (Fact_Admissions, from cleaned_dataset.csv) joined to a Dim_Date table on Date of Admission, in a one-to-many relationship. Hospital and Doctor were not split into separate dimension tables because they are near-unique per record (39,876 and 40,341 distinct values respectively out of 54,966 rows) and function as descriptive attributes rather than reusable dimensions.

Key measures: Total Patients (DISTINCTCOUNT of Name), Total Billing Amount (SUM), Average Billing per Patient, Average Length of Stay, Total Admissions, Abnormal Test Result %, and Emergency Admissions %. Full DAX definitions are documented in PowerBI/Data_Model_and_DAX_Build_Guide.md.

5. ANALYSIS & INSIGHTS

- **Total Patients:** 40,240 unique patients across 54,966 admission records
- **Total Billing Amount:** \$1.40 billion, with an average of \$25,550 billed per patient
- **Average Length of Stay:** 15.50 days across all admissions
- **Abnormal Test Result Rate:** 33.54% of admissions return an Abnormal result, closely followed by Normal (33.35%) and Inconclusive (33.11%) — an almost perfectly even three-way split
- **Admissions by Type:** Elective (33.61%), Urgent (33.46%), and Emergency (32.93%) are nearly identical in share, indicating a balanced mix of planned and unplanned care
- **Admissions Over Time:** Volume rose sharply from 7.3K (2019, partial year) to a steady 10.8K-11.2K per year through 2020-2023, before dropping to 3.8K in 2024 (also a partial year, as the data ends 07-05-2024) the dip reflects incomplete-year data, not a real decline
- **Medical Conditions:** All 6 tracked conditions (Arthritis, Diabetes, Hypertension, Obesity, Cancer, Asthma) are nearly identical in volume, each accounting for roughly 9.1K-9.2K admissions (~16-17% each)
- **Gender:** Nearly an even split, 27.5K Male vs 27.5K Female patients
- **Insurance Billing:** All 5 providers bill within a tight range — Cigna (\$284M), Medicare (\$283M), Blue Cross (\$280M), United Healthcare (\$280M), Aetna (\$277M) no provider stands out as disproportionately costly
- **Top Hospitals by Volume:** LLC Smith leads with 44 admissions, followed by Ltd Smith (39), Smith Ltd and Johnson PLC (37 each), and Smith PLC / Smith Group (36 each)

6. CONCLUSIONS

The finished dashboard confirms what the earlier assessment suggested: this is a remarkably balanced dataset. No single medical condition, admission type, gender, or insurance provider dominates the numbers, and the only notable pattern — the drop in 2024 admissions — is explained by the dataset's date range ending mid-year rather than a genuine operational issue. Data quality was strong after cleaning (534 duplicates removed, 108 negative billing values corrected), and the resulting dashboard delivers reliable, decision-ready KPIs across patient volume, billing, and clinical outcomes.

7. RECOMMENDATIONS

- Investigate the root cause of the 108 originally-negative Billing Amount records with the source system owner before relying on billing totals for financial reporting
- When reviewing the Admissions Over Time chart, note that 2019 and 2024 are partial years (data starts 08-05-2019 and ends 07-05-2024) — exclude or annotate these when comparing year-over-year trends
- Set a monitoring threshold on Abnormal Test Result % (currently 33.54%) to flag any upward drift for clinical review
- Use Average Length of Stay (15.50 days) as a recurring operational KPI to identify departments or conditions driving longer stays
- On the Overview page, replace the duplicated 'Average Length of Stay' card with a distinct metric (e.g. Total Admissions) to avoid redundant KPIs
- Consider enriching future data collection with a dedicated Hospital ID and Doctor ID (rather than free-text names) to enable proper dimension modeling in later iterations

8. DASHBOARD OVERVIEW

The final Power BI dashboard (PowerBI/pbix/dashboard.pbix) spans 3 pages: Overview (KPI summary, admissions trend, admission type mix), Demographics & Admissions (medical condition breakdown, gender split, top hospitals table), and Billing, Insurance & Clinical Outcomes (billing by provider, test result outcomes). Filters for Date, Medical Condition, Gender, and Insurance Provider apply consistently across all pages.

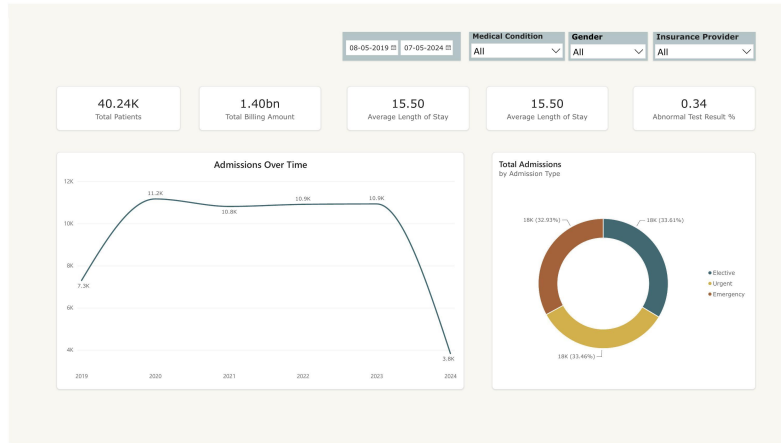


Figure 1: Overview page

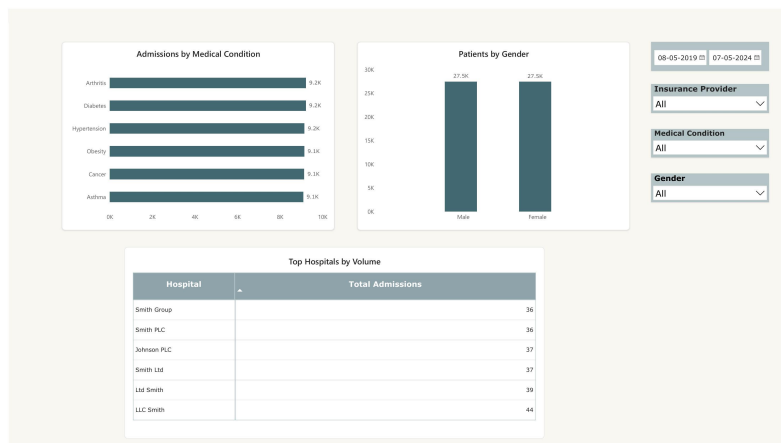


Figure 2: Demographics & Admissions page

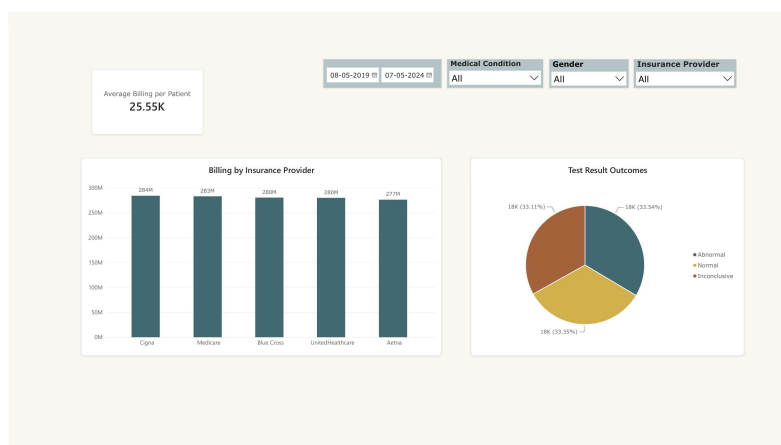


Figure 3: Billing, Insurance & Clinical Outcomes page

9. NOTES / LIMITATIONS

- Dataset is synthetic (Kaggle “Healthcare Dataset”) and does not reflect a real hospital system; insights are illustrative of the analysis approach rather than actionable real-world findings
- The correction of negative Billing Amounts to absolute values is an assumption documented in the BRD; the true cause (refunds vs. data entry error) is unknown
- Hospital and Doctor fields could not be modeled as clean dimensions due to near-unique cardinality relative to row count
- 2019 and 2024 are partial years in the dataset, which affects year-over-year trend comparisons on the Admissions Over Time chart
- The Overview page currently shows two ‘Average Length of Stay’ cards; one should be swapped for a distinct metric in the next revision